Geavanceerde computerarchitectuur: lab-4

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1 Learning objective

In this lab session we learn about allocating memory in the GPU asynchronously. We will try to see how much faster this would run compared to synchronous allocation.

2 Experiment

In this lab we will be using some simple operations for an array such as add, multiply, min and max operations. These will each need a separate array so we can compare execution speeds between doing this synchronously and asynchronously. We will be using known tactics from previous labs such as reduction and shared memory access. The expected outcome will be a big difference between the execution speeds especially when the array gets bigger. Whilst the synchronous method will need to wait for the array to be allocated (which will take longer the bigger the array is) the asynchronous method can continue. This is very visible in the results. The asynchronous way is better in all cases because of the before mentioned method and it only increases a small bit whilst the synchronous method increases in a linear way. For this lab I made two functions which each executed one of the methods 1000 times and took the average.

3 Conclusions

When optimising your Cuda program you can drastically increase your executing speeds by using asynchronous allocation. This is because the asynchronous method doesn't make the program wait until its finished. The bigger the array or element you want to use in Cuda the better it is to implement asynchronous allocation.

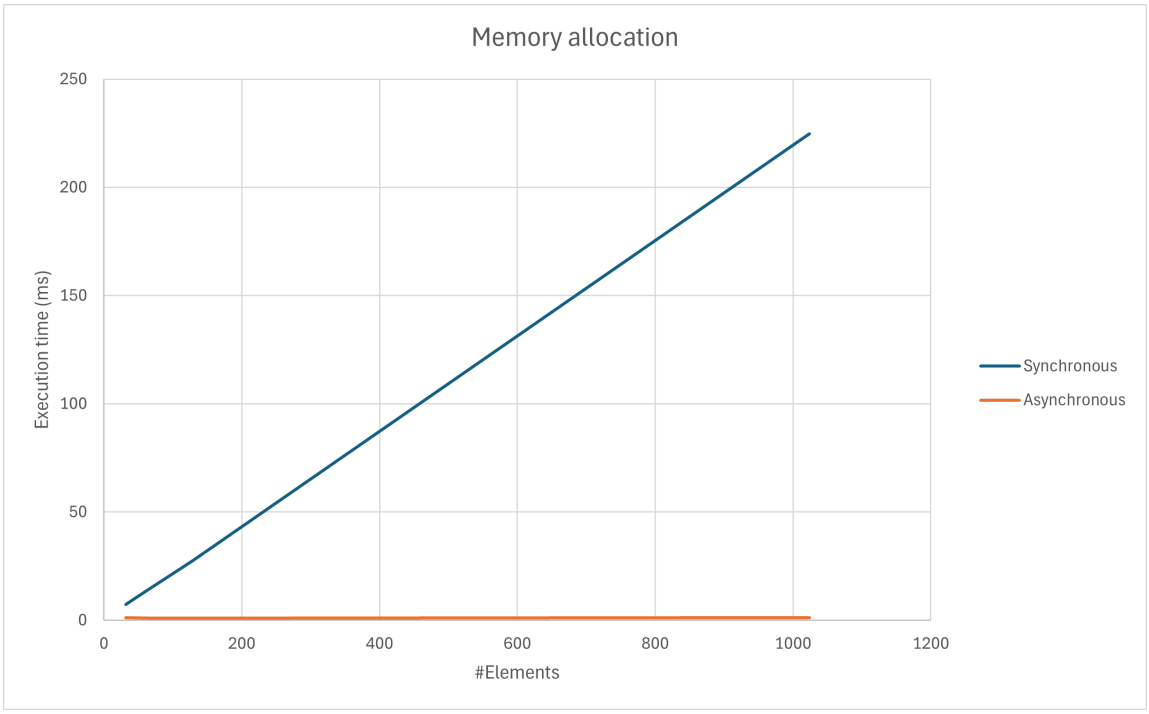


Figure 1: executing speeds for asynchronous and synchronous memory allocation